

A tensor conductivity law that removes the first-order diffuse-interface bias from homogenized $\mathbf{K}_{\text{eff}}$

1 The problem

The effective conductivity of a periodic ice/air cell is obtained from the corrector problem

$$\nabla \cdot (\mathbf{K}(\nabla t_m + \mathbf{e}_m)) = 0 \text{ in } \Omega, \quad t_m \text{ periodic}, \quad \mathbf{K}_{\text{eff}} \mathbf{e}_m = \langle \mathbf{K}(\nabla t_m + \mathbf{e}_m) \rangle, \quad (1)$$

for each direction m . With a sharp interface, $\mathbf{K} = K_\star \mathbf{I}$, where $K_\star = K_i$ in ice and K_a in air; this is the problem of Calonne et al. (2011). With a phase field φ the coefficient must also be defined *inside* the diffuse band, and the usual choice is the arithmetic law $K_{\text{ar}}(\varphi) = K_a + (K_i - K_a)\varphi$. The two problems coincide exactly when φ is the indicator function of the ice (companion note, Prop. 1). At finite ε they differ, by an amount of order ε that depends on the interpolation law. For $K_i/K_a = 114.5$ this bias is large: +17% at $\varepsilon = L/128$ for a laminate loaded normal to its layers.

The interpolation is invisible to the sharp problem, since every law with $K(0) = K_a$, $K(1) = K_i$ agrees on the indicator. It is therefore free to choose, and we choose it to cancel the first-order bias.

2 Structure of the diffuse band

Let s be the signed distance to the interface Γ (the $\varphi = \frac{1}{2}$ level set), positive in ice, and let the profile be logistic,

$$\varphi = \sigma(s/\varepsilon), \quad \sigma(u) = \frac{1}{1 + e^{-u}}, \quad \sigma(-u) = 1 - \sigma(u). \quad (2)$$

When ε is small compared with the interface's radius of curvature and with the spacing between interfaces, the band is locally a flat one-dimensional layer. Write the temperature gradient $\mathbf{E} = \nabla T$ and the flux $\mathbf{q} = \mathbf{K}\mathbf{E}$. At leading order in the stretched coordinate $u = s/\varepsilon$, two quantities are constant across the layer:

- the *tangential gradient* $\mathbf{E}_t$, because $\mathbf{E}$ is a gradient (curl-free), so its tangential part cannot vary over the short normal distance $O(\varepsilon)$;
- the *normal flux* $q_n = \mathbf{q} \cdot \hat{\mathbf{n}}$, because $\nabla \cdot \mathbf{q} = 0$ and the tangential flux divergence is $O(1)$ while ∂_s is $O(\varepsilon^{-1})$.

These are exactly the quantities that the sharp interface conditions make continuous: the temperature (hence $\mathbf{E}_t$) and the normal flux. Inside the band, then, the tangential sub-layers act in *parallel* and the normal ones in *series*.

3 Surface excesses and the first-order bias

Because the band acts in parallel along Γ and in series across it, it is equivalent, to first order, to a sharp interface carrying two excesses per unit area:

$$\Sigma_t = \int_{-\infty}^{\infty} [K_t(\varphi) - K_\star] ds, \quad \Sigma_n = \int_{-\infty}^{\infty} \left[\frac{1}{K_n(\varphi)} - \frac{1}{K_\star} \right] ds, \quad (3)$$

where K_t and K_n are the band's conductivities along and across the interface ($K_t = K_n = K$ for a scalar law). Σ_t is an excess *conductance*, Σ_n an excess *resistance*. This is the imperfect-interface (thin-interphase) limit of Hashin (2001) and Benveniste (2006).

Proposition 1. *For a unit mean gradient along $\mathbf{e}$,*

$$\mathbf{e} \cdot \mathbf{K}_{\text{eff}}^\varepsilon \mathbf{e} - \mathbf{e} \cdot \mathbf{K}_{\text{eff}}^\star \mathbf{e} = \frac{1}{|\Omega|} \int_\Gamma \left(\Sigma_t |\mathbf{E}_t|^2 - \Sigma_n q_n^2 \right) d\Gamma + O(\varepsilon^2), \quad (4)$$

with $\mathbf{E}_t$ and q_n taken from the sharp solution.

Justification. $\mathbf{e} \cdot \mathbf{K}_{\text{eff}} \mathbf{e}$ is a minimum of the energy $\langle \mathbf{E} \cdot \mathbf{K} \mathbf{E} \rangle$ (Dirichlet principle) and, dually, a maximum in terms of the flux (Thomson principle). Split the band's energy density as $K_t E_t^2 + q_n^2/K_n$. A change in tangential conductance at fixed $\mathbf{E}_t$ alters the energy by $\Sigma_t E_t^2$. A change in normal resistance at fixed q_n alters the complementary energy by $\Sigma_n q_n^2$, which changes the conductance by $-\Sigma_n q_n^2$. By stationarity, the accompanying change in the field enters only at second order. The companion note (Eq. 21) gives a rigorous two-sided bound of the same quantity that does not rely on the expansion.

Check (laminate). For layers of period L with n_Γ interfaces, a gradient normal to the layers has $\mathbf{E}_t = 0$ and $q_n = k_\perp$, and (4) gives $\delta \langle K^{-1} \rangle = \Sigma_n n_\Gamma / L$, exactly the closed form. A gradient along the layers has $q_n = 0$, $|\mathbf{E}_t| = 1$, and gives $\delta k_\parallel = \Sigma_t n_\Gamma / L$.

4 Excesses of the scalar laws

Substituting (2), $p = \sigma(u)$, $du = dp/[p(1-p)]$ and partial fractions gives closed forms (checked by quadrature):

law	Σ_t (excess conductance)	Σ_n (excess resistance)
arithmetic, $K_{\text{ar}} = K_{\text{a}} + (K_{\text{i}} - K_{\text{a}})\varphi$	0	$-\varepsilon (K_{\text{a}}^{-1} - K_{\text{i}}^{-1}) \ln(K_{\text{i}}/K_{\text{a}})$
harmonic, $K_{\text{h}}^{-1} = \varphi/K_{\text{i}} + (1-\varphi)/K_{\text{a}}$	$-\varepsilon (K_{\text{i}} - K_{\text{a}}) \ln(K_{\text{i}}/K_{\text{a}})$	0

The zeros have one cause. If a quantity is *affine* in φ , its excess is $(\text{jump}) \cdot \varepsilon \int [\sigma(u) - H(u)] du$, and the integrand is odd by the antisymmetry in (2), so the integral vanishes. K_{ar} is affine, so $\Sigma_t = 0$; K_{h}^{-1} is affine, so $\Sigma_n = 0$. The nonzero entries come from the *other* quantity, which is not affine: $1/K_{\text{ar}}$ and K_{h} are both convex in φ , so each lies below the chord joining its end values, and both excesses are negative.

Physically, the arithmetic law short-circuits the series resistance, so the cell conducts too well across interfaces. The harmonic law blocks the parallel path, so it conducts too poorly along them. At $K_{\text{i}}/K_{\text{a}} = 114.5$ the two nonzero constants are $234.96 \varepsilon \text{ m K W}^{-1}$ and $10.76 \varepsilon \text{ W m}^{-1} \text{ K}^{-1}$. No scalar law can null both, because that would require K and $1/K$ to be affine in φ at once.

5 The tensor law

Use each law in the direction where its excess vanishes:

$$\mathbf{K}(\varphi, \nabla \varphi) = K_{\text{ar}}(\varphi) (\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_{\text{h}}(\varphi) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}, \quad \hat{\mathbf{n}} = \frac{\nabla \varphi}{|\nabla \varphi|}. \quad (5)$$

Then $K_t = K_{\text{ar}}$ and $K_n = K_{\text{h}}$ in (3), so

$$\Sigma_t = \Sigma_n = 0, \quad \mathbf{K}_{\text{eff}}^\varepsilon - \mathbf{K}_{\text{eff}}^\star = O(\varepsilon^2). \quad (6)$$

Properties.

1. *Same sharp problem.* At $\varphi \in \{0, 1\}$ both branches equal $K_\star$, so $\mathbf{K} = K_\star \mathbf{I}$ in the bulk and the sharp limit is untouched (companion note, Lemma 1). Only the band changes.
2. *Well posed.* $\mathbf{K}$ is symmetric with eigenvalues K_{ar} (tangential) and K_{h} (normal), both in $[K_{\text{a}}, K_{\text{i}}]$. The cell problem remains uniformly elliptic with the same bounds, and its discrete operator remains symmetric positive definite.
3. *No free parameters.* Nothing is fitted or calibrated. The law follows from the two excess conditions alone and holds for any $K_{\text{i}}/K_{\text{a}}$.
4. *Undefined normal is harmless.* $\hat{\mathbf{n}}$ is undefined where $\nabla\varphi = 0$, which happens in the bulk, where $K_{\text{ar}} = K_{\text{h}}$ and $\mathbf{K}$ is isotropic whatever $\hat{\mathbf{n}}$ is. We set $\mathbf{K} = K_{\text{ar}} \mathbf{I}$ there.

6 Why this is legitimate

The corrector problem (1) is post-processing of a *frozen* microstructure. The phase-field evolution (vapour transport, latent heat, interface kinetics) is solved with its own interpolation and is unaffected. The two interpolations answer different questions. The evolution law sets how the geometry develops; the homogenization law sets how a given geometry is turned into $\mathbf{K}_{\text{eff}}$. Only the second is changed. The same laws already give the same sharp problem, so the change amounts to choosing, among the band representations of that problem, the one with no first-order error.

This parallels the thin-interface corrections of Karma & Rappel (1998), Almgren (1999) and Kaempfer & Plapp (2009), which remove $O(\varepsilon)$ artifacts from phase-field *dynamics* (for example, with an anti-trapping current or a renormalized kinetic coefficient). Here the problem is static and has no source term, so the thin-interface analysis stops at the two equilibrium excesses (3), and it closes in elementary form. The comparison is one of method only. The tensor law is a property of the steady-state homogenization, and it is not an argument for changing the energy balance of any metamorphism model.

7 Verification

Two cells with sharp closed forms, run over an ε ladder at fixed $\varepsilon/h = 5.12$ (so the interface bias is not confounded with mesh convergence):

1. *Laminate* ($\bar{\varphi} = \frac{1}{2}$, $n_\Gamma = 2$). Along the layers $k_\parallel = \langle K_{\text{ar}} \rangle$ is exact for both laws. Across the layers $1/k_\perp = \langle 1/K_n \rangle$ is linear in ε with slope $(n_\Gamma/L) |\Sigma_n|/\varepsilon$ for the arithmetic law, and exactly flat at the sharp value for the tensor law. The laminate's normal is a fixed coordinate axis, so this case does not exercise the off-diagonal part of (5).
2. *Disk array* (square array, $f = \pi R^2/L^2 = 0.196$). The sharp value is Rayleigh's, $k = 0.0295684 \text{ W m}^{-1} \text{ K}^{-1}$ (Perrins et al. 1979); Maxwell-Garnett agrees to 1.8×10^{-4} . Here the normal takes every direction. Using the dipole field in (4) gives the arithmetic law's slope with no fitted constants,

$$\frac{dk}{d\varepsilon} = \frac{|\Sigma_n|}{\varepsilon} \frac{\pi R}{L^2} \left[\frac{K_{\text{a}}(1 + \beta)}{1 - \beta f} \right]^2, \quad \beta = \frac{K_{\text{i}} - K_{\text{a}}}{K_{\text{i}} + K_{\text{a}}}, \quad (7)$$

which is +15% at $\varepsilon = L/100$ and +1.9% at $L/800$. The disk ladder starts at $L/100$: at $L/50$, $\varepsilon/R = 0.08$ and the visible band spans most of the radius, outside the thin-interface regime (measured there: arithmetic +46%, tensor +6.7%).

ε	laminate, $k_{\perp}$ bias	
	arithmetic (pred.)	tensor (meas.)
$L/50$	+59.4%	$< 4 \times 10^{-7}$
$L/64$	+41.1%	$< 4 \times 10^{-7}$
$L/128$	+17.0%	$< 4 \times 10^{-7}$
$L/256$	+7.9%	$< 4 \times 10^{-7}$
$L/512$	+3.8%	$< 4 \times 10^{-7}$

On the laminate the tensor law is exact to solver precision at every rung, as predicted. The fitted slope of $1/k_{\perp}$ is 0.10% of the arithmetic law's.

ε	disk array, k bias			
	arith. (pred.)	arith. (meas.)	tensor (meas.)	tensor, IC offset removed
$L/100$	+15.07%	+18.46%	+1.377%	+1.219%
$L/200$	+7.53%	+8.37%	+0.283%	+0.205%
$L/400$	+3.77%	+3.99%	+0.073%	+0.034%
$L/800$	+1.88%	+1.95%	+0.026%	+0.006%

For the arithmetic law, a quadratic fit in ε gives a first-order coefficient of $440.9 \text{ W m}^{-2} \text{ K}^{-1}$ against the predicted 445.5 (−1.0%). The intercept is within 0.04% of Rayleigh's value. This confirms Prop. 1 on a curved interface with no fitted constants. For the tensor law the first-order coefficient is −2.5% of the arithmetic one, and at every rung the error is at most $0.075\times$ the arithmetic error. The last column removes an offset in the benchmark's initial condition, which stretched the disk by $(N+1)/N$ on an N^2 mesh (measured area excess $2/N$ to four digits). It has since been fixed in the code, so a rerun measures this column directly. With the offset removed, the tensor residual falls by about $6\times$ per halving of ε (observed order ≈ 2.5), consistent with (6).

Drivers: `enceladus_DSM/studies/keff_sharp_limit/verification/ (--interp tensor)` and `.../disk/`.

8 Scope and limitations

- *Residual* $O(\varepsilon^2)$. This includes curvature terms and a geometric one: a diffuse disk holds $\pi^3 \varepsilon^2/3$ more area of ice than the sharp disk.
- *Thin ligaments*. The derivation assumes the band is a flat 1D layer, which fails where two interfaces are closer than the band width (about $6-9\varepsilon$), as in young sinter necks. No interpolation law is justified there. This limits how well the phase field resolves the geometry, and it applies to the evolution model's geometry as much as to $\mathbf{K}_{\text{eff}}$.
- *Profile*. The zero excesses use the antisymmetry of the logistic profile. Any profile with $\varphi(-s) = 1 - \varphi(s)$ about the $\varphi = \frac{1}{2}$ level set gives the same result; a profile distorted by the evolution changes it only at the next order.
- *Geometry error is untouched*. The law removes the bias of homogenizing a given φ . It does not correct errors in φ itself.

9 Suggested Methods text

The effective conductivity tensor is obtained by periodic homogenization (Calonne et al., 2011), solving the corrector problem on the frozen phase field at each sampled time. Inside the diffuse interface the conductivity must be interpolated. A scalar interpolation carries a first-order bias in the interface width ε : to first order the band behaves as a sharp interface with an excess tangential conductance Σ_t and an excess normal resistance Σ_n (Hashin, 2001; Benveniste, 2006). For the logistic profile, the arithmetic law gives $\Sigma_t = 0$ but $\Sigma_n = -\varepsilon(K_a^{-1} - K_i^{-1}) \ln(K_i/K_a)$, which overestimates conduction across interfaces by up to tens of percent at $K_i/K_a \approx 115$. The harmonic law has the opposite defect. We therefore use the tensor law $\mathbf{K} = K_{\text{ar}}(\varphi)(\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_{\text{h}}(\varphi) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}$, $\hat{\mathbf{n}} = \nabla\varphi/|\nabla\varphi|$, whose excesses both vanish, so the error is $O(\varepsilon^2)$. It reduces to the sharp conductivities in each bulk phase, has no adjustable parameters, and affects only the homogenization step, not the phase-field evolution. We verified it against closed-form sharp solutions for a laminate and for a square array of disks (Rayleigh, 1892; Perrins et al., 1979).

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